Bayesian Deferral for Seismic Phase Picking

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Background and Motivation

- **Deep learning models:** allow automated phase picking in Seismology
  - Phase picking is the critical first step for earthquake location
- **The Silent Failure:** When deployed in data-scarce regions like Kazakhstan, models encounter unfamiliar waveforms (intraplate tectonics, different instruments).
- **The Problem:** no signal that the input is out-of-distribution
- **Goal:** A model that knows when it is uncertain and defers decisions to humans when it is uncertain

Data and Model

- PhaseNet
  - U-Net designed to pick earthquake phases
  - Input: 3-component ground motion, sampled as time-series
  - Output: Probability of P-waves, S-waves, Noise for each timestamp in the input window
- Northern California (NCSN)
  - Training domain
  - Dense network (800 stations) and a specific strike-slip tectonic regime

- Kazakhstan
  - Sparse network (13 primary stations)
  - Geology and earthquake types (intraplate) are fundamentally different

Non-stationarity

- In seismology, the statistical properties of the data (e.g. event frequency and waveform character) change over time.
- **Reliability Drift:** PhaseNet's accuracy is not a static value, depends on seismic environment
  - High during quiet periods
  - Drops sharply during active periods, environment shifted away from training baseline
- To manage non-stationarity, model the environment as a sequence of three hidden states:
  - **Regime Q (Quiet):** Background activity, standard waveforms. High reliability
  - **Regime A (Active):** High-magnitude events. Low reliability due to waveform complexity and signal saturation
  - **Regime D (Decaying):** The transition phase, aftershock waves decrease, and the environment is slowly returning to normal
- By tracking these regimes, the system can dynamically adjust its trust in the model rather than relying on a fixed, global confidence threshold

Methodology

Phase 1: Baseline Performance (In-Distribution)

- Run pre-trained PhaseNet on Northern California Dataset

Phase 2: Distribution Shift and Uncertainty Quatification

- Run pre-trained PhaseNet on Kazakhstan Dataset
- Compare Uncertainty Quantification Methods:
  - Baseline: Built-in confidence from PhaseNet
  - MC dropout: Perform T forward passes with dropout and measure variance
- Quantify the gap to justify deferral

Phase 3: Regime Modelling

- **Goal:**
  - The goal of this recursive Bayesian process is to dynamically track the non-stationary seismic environment by updating regime beliefs at each time step based on both history and new earthquake observations.

  - By calculating a **regime posterior** that quantifies the probability of being in each regime, the system can estimate PhaseNet's current reliability based on this probability and decide whether to defer high-stakes decisions to a human expert to prevent silent failures.

Phase 3: Regime Modelling

Goal: Implement the Forward Algorithm to estimate the current Regime

- **3.1. Feature Engineering:** Extract observations for each time window: total event count and max magnitude
- **3.2. Parameter Estimation**
  - Define Poisson rates to quantify expected activity for each regime (using observations)
  - e.g. Active regime: expecting 50 earthquakes per time window
- **3.3. Forward Algorithm:** Implement the recursive Bayesian update
  - **Predict:** Calculate Prior at t = Prior at t-1 * Transition Probability
    - Predict the current regime based on our belief from previous time window & how earthquakes usually transition
  - **Update:** Probability Score = Likelihood (Poisson) * Prior
    - Look at current observations and check how well it fits Quiet, Active, Decaying profiles; Combine this with Prior
  - **Normalize:** Regime Posterior = Probability Score / Sum of Probability Scores
- **3.4. Validation:** Check if the posterior correctly identifies known historical earthquake sequences in Kazakhstan

Phase 4: Bayesian Reliability Updates

Goal: Track regime-specific model reliability using **Beta-Bernoulli** conjugate priors.

- **4.1. Beta Prior Initialization:** Set Beta params for each regime
- Starting from weak priors
- **4.2. Reliability Calculation:** Compute expected reliability at time-stamp

$$\hat{\theta}_t = \sum \pi_t(r) \cdot \frac{\alpha_r}{\alpha_r + \beta_r}.$$

Expected Reliability of PhaseNet in that Regime

Probability in Specific Regime (Regime Posterior)

4.3. Compare weighted reliability to safety threshold

If weighted reliability < threshold: Defer

- **4.4.Soft-Update Mechanism:** After pick is verified:

$$\alpha_{r,new} = \alpha_{r,old} + (1 \cdot \pi_t(r))$$ If correct

$$\beta_{r,new} = \beta_{r,old} + (1 \cdot \pi_t(r))$$ If incorrect

Updating expected reliability for each regime

Assumptions

- **A1.1: Non-stationarity**: The seismic environment is non-stationary - the probability distribution governing seismic activity changes over time.
- **A1.2: Regime approximation**: Non-stationarity is modelled as transitions between three discrete regimes (Quiet, Active, Decaying).
- **A2.1: First-order Markov**: Regime transitions depend only on the current regime, not the full history.
- **A3.1: Uncertainty growth after events in data-sparse regions**: PhaseNet becomes less reliable after significant seismic activity in Kazakhstan, because post-event waveforms are further from the California training distribution.
- **A4.1: Noisy observation**: PhaseNet's output is treated as a noisy observation of pick quality, not ground truth.
- **A4.2: Beta priors**: Regional reliability follows a Beta distribution with regime-specific parameters.
- **A4.3: Conditional independence**: Picks within a time window are independent given the regime.
- **A5.1: Expert superiority**: The human analyst is more accurate (95%) but capacity-limited (20–30% of picks).
- **A5.2: Constant deferral cost**: Each deferral incurs the same cost regardless of context.
- **A5.3: Binary deferral**: The decision is defer or don't -no partial deferral or graded confidence flags.